Curtin University
School of Electrical Engineering,
Computing & Mathematical Sciences

TESSELE

Our Priority SDGs

Powering T&L with Waste-to-Energy Systems:

A Dynamic Optimisation & Machine Learning Framework

TEAM 1
THE GREEN ALGORITHMS
Data. Insight. Impact.

An integrated, data-driven approach to advance sustainability education and innovation, combining synthetic data, machine learning, and optimisation to solve real-world energy challenges

Synthetic Data Generation Framework

Create realistic, scalable, and reusable data for sustainability experiments.

+

Machine Learning Framework

Build predictive models to forecast energy, waste, and resource outcomes.

+

Optimisation Framework

Develop dynamic optimisation to improve resource efficiency and minimise emission.

Curriculum Integration

Embed the framework into teaching through case studies, assignments, and projects.

Phase 1
Data Collection & Exploration
12 May

Phase 2
Data Engineering
1 June

Phase 3
ML Framework
1 July

Phase 4
Assignments & Project Design
1 Aug

Phase 5
Optimisation Framework
1 Sep

Phase 6
Evaluation of Curriculum Integration
1 Oct

Final Showcase & Recognition Event
5 Nov

Data-driven. Sustainability-focused. Education-powered.
Building a smarter, greener future through innovation and collaboration.

CRICOS Provider Code 00301J